Supplementary Material

Scan Once, Localize Anywhere: Real-Time Monocular Localization in Gaussian Splatting Maps on Edge Devices

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This document tabulates two numerical grids summarised in the main paper, details the held-out evaluation of the surveyed grid cited there, and reports a synthetic motion-blur test. Every value was produced by the same frozen pipeline configuration described there (ALIKED `aliked-n32`; FAISS exact flat L_2; PnP-RANSAC, `SOLVEPNP_ITERATIVE`, 2.5 px, confidence 0.9999, 500 iterations; failures reported, never substituted).

S1 Complete stability-filter grid (70 configurations)

Seven 7-Scenes scenes × two splat representations × five filter settings. Each cell gives **landmarks retained / 5 cm 5° success rate (%)**. The notation $\ell N/M$ denotes lifetime $\geq N$ observations and $\|\sigma_{xyz}\| \leq M$ mm; the descriptor gate is held at 0.28 throughout. `raw` is the unfiltered database and $\ell 5/5$ is the deployed setting behind every headline number. Success follows the paper's failure convention: frames the pipeline declines to localize count as infinite error and remain in the denominator.

(a) 2DGS-regularized maps

Scene	raw	$\ell 3/10$	$\ell 5/10$	$\ell 3/5$	$\ell 5/5$ (deployed)
Chess	143,744 / 94.8	36,985 / 96.0	25,696 / 96.0	28,236 / 95.7	19,495 / 95.1
Fire	118,567 / 63.0	28,419 / 63.7	22,111 / 62.7	23,731 / 64.5	19,491 / 64.6
Heads	95,082 / 24.5	9,223 / 30.0	2,469 / 28.4	8,030 / 26.5	1,784 / 24.2
Office	419,448 / 71.7	45,992 / 74.2	30,020 / 73.4	32,061 / 72.2	21,076 / 71.7
Pumpkin	56,708 / 81.0	17,553 / 81.4	13,337 / 80.5	11,508 / 80.5	8,438 / 79.8
RedKitchen	381,272 / 82.0	63,738 / 81.6	40,196 / 81.1	50,444 / 80.0	31,653 / 79.5
Stairs	352,659 / 7.9	31,690 / 3.4	11,367 / 3.5	18,810 / 2.8	6,637 / 3.0

(b) Photometric 3DGS maps

Scene	raw	$\ell 3/10$	$\ell 5/10$	$\ell 3/5$	$\ell 5/5$ (deployed)
Chess	89,146 / 78.8	17,017 / 69.2	10,705 / 63.6	8,632 / 60.9	5,135 / 53.8
Fire	113,501 / 62.5	22,467 / 53.9	16,447 / 51.0	10,909 / 40.9	7,509 / 36.6
Heads	82,941 / 63.2	9,081 / 57.2	5,296 / 57.5	4,477 / 54.0	2,526 / 51.2
Office	316,200 / 48.3	28,122 / 39.9	14,270 / 35.4	11,309 / 32.1	4,769 / 24.6
Pumpkin	45,745 / 64.6	6,408 / 57.5	3,865 / 51.5	2,245 / 39.2	1,110 / 33.3
RedKitchen	129,593 / 58.6	31,796 / 47.2	17,726 / 43.5	15,871 / 33.7	7,179 / 27.5
Stairs	90,341 / 14.5	13,937 / 3.0	5,368 / 2.3	5,108 / 0.3	1,688 / 0.4

Notes. (i) Full-pipeline survival quoted in the paper is $\ell 5/5$ landmarks divided by `raw` landmarks; e.g. RedKitchen 2DGS 31,653/381,272 = 8.3%. (ii) On the five non-degenerate scenes the *minimum* 2DGS success across all five settings exceeds the *maximum* 3DGS success; the margin is narrowest on Fire (62.75% vs. 62.55%), where the two coincide to within run-to-run variation. (iii) RANSAC is unseeded, so repeated runs vary by roughly ±0.3 percentage points of success rate. (iv) Median translation errors for the 2DGS maps at the `raw` and $\ell 5/5$ settings are given in Table 1 of the main paper; the three intermediate gates and the 3DGS medians are not tabulated. S1 gives the success-rate half of the frontier.

S2 Surveyed-grid frame recovery: per-marker fit residuals

The rigid splat-to-world transform used in the surveyed-grid experiment is recovered by the Kabsch algorithm (rotation and translation, *no* scale parameter) from 14 marker correspondences. Surveyed coordinates were placed with a laser range finder; splat-frame coordinates were obtained by annotating each marker centre in

rendered views and back-projecting through rendered depth. The correspondences are independent of the query localizations.

Marker	Surveyed position (m)	Splat-frame position (m)	Fit residual (mm)
0	(0.000, 0.000, 0.000)	(0.060, -0.010, -0.191)	7.9
1	(-1.004, 0.000, 0.000)	(-0.940, -0.019, -0.194)	5.8
2	(-2.001, 0.000, 0.000)	(-1.940, -0.022, -0.201)	6.4
3	(-3.001, 0.000, 0.000)	(-2.938, -0.025, -0.209)	6.1
4	(-3.999, 0.000, 0.000)	(-3.937, -0.032, -0.211)	6.1
5	(-3.999, -0.501, 0.000)	(-3.931, -0.530, -0.211)	4.4
6	(-3.999, -1.000, 0.000)	(-3.924, -1.032, -0.208)	3.0
7	(-3.999, -1.501, 0.000)	(-3.917, -1.531, -0.208)	7.4
8	(-2.999, -1.502, 0.000)	(-2.916, -1.525, -0.207)	6.7
9	(-1.999, -1.503, 0.000)	(-1.917, -1.522, -0.206)	6.5
10	(-0.999, -1.501, 0.000)	(-0.917, -1.517, -0.200)	6.1
11	(0.001, -1.502, 0.000)	(0.083, -1.510, -0.200)	8.2
12	(0.001, -1.002, 0.000)	(0.075, -1.012, -0.189)	5.0
13	(0.000, -0.501, 0.000)	(0.069, -0.511, -0.189)	4.4
		mean / max / std	**6.0 / 8.2 / 1.4**

Notes. (i) All markers lie on the floor plane, $z = 0$ by construction in the surveyed frame. (ii) The recovered transform contains a 19.2 cm vertical translation—the height of the scanner's origin above the floor—and a 0.45° rotation, mostly about the vertical (0.36°); its out-of-plane component is 0.27°. (iii) The 14 splat-frame markers fit a common plane to 3.0 mm RMS (6.1 mm max), and the fit residual shows no significant trend with position ($r = +0.26$ in x, -0.13 in y; $n = 14$), indicating scatter rather than a systematic map warp. (iv) These are *fit* residuals of the frame recovery, not per-marker localization errors.

S3 Surveyed-grid held-out evaluation

In the main surveyed-grid result the same 14 markers both fit the splat-to-survey transform and score localization. Here every scored marker is held out of the fit that scores it. Only the transform changes: each held-out marker's ground-truth camera position is rebuilt with the refitted transform (marker position at the 0.88 m tripod height plus the 14.5 mm screw-to-optical-centre offset), and the localized poses of the main paper are rescored without re-localization. Errors are full 3D camera-position errors (cm); no query fails in any condition, in-sample or held out.

Database (lights on)	In-sample (paper)	Two-fold, 7/7	Leave-one-out
Pruned	2.26	2.24	2.23
Raw	2.10	2.12	2.11

Two-fold: the transform is fitted on 7 markers and scores the other 7, over all $\binom{14}{7}/2 = 1{,}716$ splits, so every marker is scored by a transform it took no part in fitting. *Leave-one-out*: each marker is scored by the transform fitted on the other 13. Per-marker leave-one-out errors for all four lighting and database conditions follow.

Marker	Pruned, lights on	Raw, lights on	Pruned, lights off	Raw, lights off
0	1.83 → 1.77	1.91 → 1.86	2.01 → 1.96	2.22 → 2.18
1	1.79 → 1.76	1.61 → 1.57	1.90 → 1.90	1.54 → 1.53
2	1.00 → 0.95	0.91 → 0.86	1.19 → 1.15	1.38 → 1.34
3	1.11 → 1.16	0.72 → 0.76	1.70 → 1.77	1.93 → 1.93
4	2.46 → 2.53	2.05 → 2.13	3.44 → 3.47	1.81 → 1.89
5	3.43 → 3.48	2.28 → 2.34	4.73 → 4.77	3.24 → 3.28
6	1.23 → 1.19	1.12 → 1.08	1.84 → 1.76	3.41 → 3.36
7	1.47 → 1.42	1.44 → 1.41	3.17 → 3.03	1.56 → 1.51
8	2.15 → 2.11	2.15 → 2.10	1.53 → 1.48	11.10 → 11.11
9	2.84 → 2.84	3.78 → 3.78	16.66 → 16.62	16.45 → 16.41
10	3.60 → 3.66	3.45 → 3.50	3.78 → 3.80	3.26 → 3.27
11	3.24 → 3.35	3.02 → 3.08	4.15 → 4.23	3.39 → 3.47
12	2.38 → 2.35	2.91 → 2.88	4.37 → 4.31	3.38 → 3.37
13	2.93 → 2.87	2.69 → 2.64	4.83 → 4.77	2.55 → 2.50
median	2.26 → 2.23	2.10 → 2.11	3.30 → 3.25	2.89 → 2.89

Notes. (i) Entries are in-sample → leave-one-out error (cm); the in-sample medians are those of the main paper. No per-marker error changes by more than 0.11 cm with the lights on or 0.15 cm with them off. (ii) The 13-marker transforms differ from the 14-marker transform by at most 0.067° and 1.1 mm, and move a held-out marker's ground truth by at most 2.6 mm (mean 1.1 mm). The 7-marker transforms of the two-fold evaluation differ by a median of 0.085° (max 0.59°) and 2.1 mm (max 18.9 mm). (iii) The large lights-off errors at markers 8 and 9 are localization errors already present in-sample; holding the marker out changes them by less than 0.05 cm.

S4 Synthetic motion blur

The main paper does not evaluate localization under motion blur. The real 7-Scenes test frames cannot isolate its effect: whether a frame localizes depends mostly on where the camera is, and the real frames contain little blur—adding even 3 px of synthetic blur leaves an image with less fine detail (variance of the Laplacian) than 89–95% of the real test frames, in every scene. We therefore blur the *same* frames synthetically, so that place, content and viewpoint stay fixed and the amount of blur is known exactly.

Method. Each raw query image is convolved with a linear motion-blur kernel, the standard synthetic motion blur of robustness benchmarks [1]: a uniformly weighted line of length L pixels through the kernel centre, so that each scene point smears into an L-pixel streak, as for a camera moving at constant velocity during the exposure. The direction is one random angle per frame (fixed seed), the same at every L, so each frame is compared only with itself. The blurred image then passes through the unchanged pipeline (undistortion, ALIKED, FAISS matching, PnP-RANSAC). We use all 17,000 7-Scenes test frames on the deployed 2DGS maps, scored at 5 cm/5°, and the 14 warehouse grid images (lights on) on the pruned and raw warehouse databases, scored on position only, since the grid has no rotation ground truth. The $L = 0$ columns report the main paper's numbers; the blurred levels were run on a different GPU.

Scene	$L = 0$	3	5	9	15	25	35 px
Chess	95.1	93.7	92.2	78.8	21.4	0.9	0.0
Fire	64.6	64.2	61.1	41.9	4.1	0.0	0.0
Heads	24.2	24.5	23.5	11.2	1.2	0.0	0.0
Office	71.7	71.2	67.6	46.3	11.0	0.9	0.3
Pumpkin	79.9	79.5	78.1	65.8	14.2	1.4	0.1
RedKitchen	79.5	78.9	76.4	58.7	16.0	0.8	0.0
Stairs	3.0	2.9	2.9	0.6	0.0	0.2	0.1
Six non-Stairs, pooled	74.2	73.7	71.2	53.9	12.8	0.8	0.1
All seven, pooled	**70.0**	**69.5**	**67.1**	**50.8**	**12.0**	**0.7**	**0.1**
Correct at $L = 0$, still correct	—	94.3	90.9	69.9	16.9	1.0	0.1

7-Scenes success (%, 5 cm/5°) by blur length. The last row gives, of the frames localized correctly without added blur in the blurred-level run, the share still correct at each L.

	Pruned database			Raw database		
L (px)	Localized	≤5 cm	Median (cm)	Localized	≤5 cm	Median (cm)
0	14	14	2.26	14	14	2.10
3	14	14	2.12	14	14	2.01
5	14	13	2.06	14	14	2.53
9	14	13	3.13	14	12	3.24
15	4	1	7.85	9	3	13.28
25	0	0	—	5	0	39.64
35	0	0	—	1	0	371.98

Warehouse grid (14 images, lights on): images localized, images within 5 cm, and median 3D position error.

	7-Scenes (640 × 480)		Warehouse (1280 × 720)		
L (px)	Keypoints	Matches	Keypoints	Matches (pruned)	Matches (raw)
0	263	37	1,188	156	339
3	228	34	1,034	129	243
5	190	29	844	92	164
9	124	13	788	40	86
15	69	0	722	0	10
25	29	0	627	0	0
35	18	0	538	0	0

Median keypoints and matches per frame (7-Scenes pooled over all 17,000 frames). A median of 0 matches means fewer than the 6 the matcher requires survived.

L (px)	7-Scenes, $f \approx 525$ px (°/s)	Warehouse, $f \approx 637$ px (°/s)
3	10–20	8–16
5	16–33	13–27
9	29–59	24–49
15	49–98	40–81
25	82–164	67–135
35	115–229	95–189

Angular speed equivalent to each blur length, $\omega = L/(f\,t_{\text{exp}})$, for an exposure t_{exp} between 33 ms (the full frame period at 30 Hz, lowest speed) and 16 ms. The exposure was automatic and is not recorded. Real 7-Scenes test frames turn at a median of 10–20°/s and a 90th percentile of 23–41°/s, depending on scene.

Notes. (i) Up to 5 px, localization is essentially unaffected: pooled 7-Scenes success goes from 70.0% to 67.1%, 91% of correct poses stay correct, and the warehouse localizes all 14 images at a median of 2.0–2.5 cm. (ii) At 9 px, 7-Scenes degrades to 50.8% success, with 70% of correct poses still correct; the warehouse, at higher resolution and with larger maps, still localizes all 14 images (median 3.1–3.2 cm; 12–13 within 5 cm). (iii) At 15 px localization collapses: 7-Scenes falls to 12.0%, the pruned warehouse database localizes 4 of 14, and the raw database localizes 9 but mostly wrongly (median 13.3 cm). (iv) Blur removes matches faster than keypoints: from 0 to 9 px, 7-Scenes keypoints roughly halve while matches fall by two-thirds, and on the pruned warehouse database keypoints fall by a third while matches fall by three-quarters. ALIKED still detects corners in blurred images, but their descriptors no longer match a map built from sharp renders. (v) Caveats: the kernel is spatially uniform and linear, which matches camera rotation but not depth-dependent translation blur, curved trajectories or rolling shutter; it is applied on top of each frame's own blur, in 8-bit gamma space, with no change in exposure or noise; the speeds rest on the exposure assumption above; the warehouse test is small (14 images, position only); and Stairs, at 3.0% success without blur, is too low to read a blur effect from.

References

[1] D. Hendrycks and T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *ICLR*, 2019.